

**Department of Biomedical Engineering
The Superior University, Lahore**

Computer Programming

Experiment No.3

The Use of Input/Output Operations (Beginner to Advanced Level)

Prepared for:

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Section: _Biomedical Engineering__

Semester: _Second_

Total Marks: _____10_____

Obtained Marks: _____

Signature: _____

Date: _____

Experiment No.3**The Use of Input/Output Operations (Beginner to Advanced Level)****(Rubrics)**Name: _Mazen Abdulgani_ID: _SU91-BBETM-F25-006_**A. PSYCHOMOTOR**

Serial No.	Criteria	Allocated Marks	Unacceptable 0%	Poor 25%	Fair 50%	Good 75%	Excellent 100%	Total Obtained
1	Follow Procedures	2	0	0.5	1	1.5	2	
2	Accuracy in Output Results	4	0	1	2	3	4	
Sub-Total		6	Sub Total Marks Obtained in Psychomotor (P)					

B. AFFECTIVE

Serial No.	Criteria	Allocated Marks	Unacceptable 0%	Poor 25%	Fair 50%	Good 75%	Excellent 100%	Total Obtained
1	Lab Report	2	0	0.5	1	1.5	2	
2	Assigned Task	2	0	0.5	1	1.5	2	
Sub-Total		4	Sub Total Marks Obtained in Affective (A)					

Instructor Name: Miss Eisha NawazTotal Marks (P+A): 10

Instructor Signature: _____

Obtained Marks (P+A): _____

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The Use of Input/ Output Operations (Beginner to Advanced Level)

1 LAB OBJECTIVE(S)

- To understand the use of print statements from beginner to advance level
- To learn print statements based on complex problems and spacing mechanism

2 INTRODUCTION TO VARIABLES

2.1 VARIABLE DECLARATION IN C++

A variable declaration provides assurance to the compiler that there is one variable existing with the given type and name so that compiler proceed for further compilation.

```
int a;  
char b;  
float c;
```

2.2 VARIABLE INITIALIZATION IN C++

A variable can be initialized at the time of declaration or even after that. Basically, initialization means storing some actual meaningful data inside the variable.

```
int a=2;  
char b='x';  
float c=2.907;
```

2.3 CONSTANTS

Constants refer to fixed values that the program may not alter during its execution. These fixed values are also called **literals**.

2.3.1 Defining Constants

There are two simple ways in C++ to define constants:

Using **#define** preprocessor

Using **const** keyword

```
#define LENGTH 10  
#define WIDTH 5  
const int LENGTH = 10;  
const int WIDTH = 5;
```

2.4 LEGAL IDENTIFIERS

Regardless of which style you adopt, be consistent and make your variable names as sensible as possible. Here are some specific rules that must be followed with all identifiers.

- The first character must be one of the letters a through z, A through Z, or an underscore character (_).
- After the first character you may use the letters a through z or A through Z, the digits 0 through 9, or underscores.
- Uppercase and lowercase characters are distinct. This means Items Ordered is not the same as Items ordered.

3 INTRODUCTION TO PRINT STATEMENT

Every programming language employs a statement to print desired output on the screen. Since we are using C++, it employs COUT function to print output on screen or console.

3.1 COUT IN C++

- COUT function are inbuilt library functions in C++ programming language which are available in C++ library by default. These functions are declared and related macros are defined in “using namespace std” which is a header file in C++ language.
- We have to include “using namespace std” file as shown in below C++ program to make use of COUT library function in C++ language.

3.1.1 Syntax

The syntax of COUT function is displayed below:

```
COUT<<“”;
```

The description is as follows:

- COUT is pre-defined statement which is necessary to be written in order to print something
- The "c" in cout refers to "character" and "out" means "output".
- Hence cout means "character output".
- The cout object is used along with the insertion operator << in order to display a stream of characters.

3.1.2 Example 1

```
#include<iostream>
```

```
int main ()
```

```
{
```

```
std::cout<< “Hello World”;
```

```
return 0;
```

```
}
```

Output

```
Hello World
```

3.2 USE OF STD

- In the previous example, we saw that before COUT we used “std::”.
- C++ moved all of the functionality in the standard library into a namespace named “std” (short for standard)
- Hence, is we use “using namespace std” before our main() function, we wont have to write std:: before the cout statement

3.2.1 Example 2 (using Namespace std)

```
#include<iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
cout<< “Hello World”;
```

```
return 0;
```

```
}
```

Output

Hello World

4 ESCAPE SEQUENCES

Escape sequences are special characters used in control string to modify the format of an output. These specific characters are translated into another character or a sequence of characters that may be difficult to represent directly. For example, you want to put a line break in the output of a C++ statement then you will use “\n” character which is an escape sequence itself.

An escape sequence consists of two or more characters. For all sequences, the first character will be “\” i.e. backslash. The other characters determine the interpretation of escape sequence. For example, “n” of “\n” tells the cursor to move on the next line.

4.1.1 New Line (\n):

When a new line is necessary in the output, then this escape sequence is used. For example:

```
cout<<”COMPUTER\nSCIENCE”;
```

First of all, “COMPUTER” is printed and “\n” shifts the cursor to the next line. Then “SCIENCE” is printed on second line. A screenshot of output is shown below:

4.1.2 Tab (\t):

- A TAB is equal to eight spaces. Whenever TAB button is pressed from the keyboard, then 8 spaces are left blank. This escape sequence performs the functionality of TAB key in the output stream.
- The code given below will insert a TAB between two words.

```
cout<<"COMPUTER\tSCIENCE";
```

C++ Tab (\t)

- As you can see after printing "COMPUTER", 8 spaces are left blank and then "SCIENCE" is printed on the screen

4.1.3 Alert Bell (\a):

- This escape sequence is used to play beep during execution. For example:

```
cout<<"COMPUTER\aSCIENCE";
```
- First of all, "COMPUTER" is printed then a beep is played and after that "SCIENCE" is printed.

4.1.4 Backspace (\b):

- Whenever we want to delete a single character, we press the button "backspace" from our keyboard. The same functionality can be achieved in C++ output with this escape sequence. For example:

```
cout<<"COMPUTER\bSCIENCE";
```

C++ Backspace (\b)

- First of all, "COMPUTER" is printed and after that "\b" comes which deletes the last character i.e. "R". After that, "SCIENCE" is printed.

4.1.5 Single Quote (\'):

- To insert a single quote in the output, this escape sequence is used. Look at the code written below:

```
cout<<"\'COMPUTER SCIENCE\'";
```

- This code prints single quotes at the start and end of the string.

4.1.6 Carriage Return (\r):

- This escape sequence moves the cursor at the beginning of current line. For example:

```
cout<<"COMPUTER\rSCIENCE";
```
- First of all, "COMPUTER" is printed and after that "\r" comes which moves the cursor at the beginning of the line and "SCIENCE" is printed which overwrites the word written before.

4.1.7 Form Feed (\f)

- This escape sequence is used in printing. It is used to insert a blank paper in the printed output. For example:

```
cout<<"COMPUTER\fSCIENCE";
```

- After printing "COMPUTER", the computer will include a blank paper and then print "SCIENCE".

4.1.8 Example 3

```
#include<iostream>

using namespace std;

int main()
{
    cout<< "#####\n";
    cout<< " #\t#\n";
    cout<< " #\t#\n";
    cout<< "#####\n";
    return 0;
}
```

Output

```
#####
#      #
#      #
#####
```


5 LAB TASKS

5.1 TASK 1

Write a C++ program to print a SUPERIOR using hash (#).

Expected Output:

```
##### # # ##### ##### ##### ##### #####
# # # # # # # # # #
# # # # # # # # # #
##### # # ##### ##### ##### # # #####
# # # # # # # # # #
# # # # # # # # # #
##### ##### # ##### # # ##### # #
```

```
1 #include <iostream>
2 using namespace std;
3 int main ()
4 {
5     cout<<"##### # # ##### ##### ##### ##### #####\n";
6     cout<<"# # # # # # # # # #\n";
7     cout<<"# # # # # # # # # #\n";
8     cout<<"##### # # ##### ##### ##### # # #####\n";
9     cout<<"# # # # # # # # # #\n";
10    cout<<"# # # # # # # # # #\n";
11    cout<<"##### ##### # ##### # # ##### # #\n";
12    return 0;
13 }
```

Process exited after 0.05341 seconds with return value 0
Press any key to continue . . .

5.2 TASK 2

Write a C++ program to print the following:

Expected Output

```
*****
#      10      $
$      789     #
#      456     $
$      123     #
*****
```

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int num;
6     cout<<"*****\n";
7     cout<<"#      10      $\n";
8     cout<<"$      789     #\n";
9     cout<<"#      456     $\n";
10    cout<<"$      123     #\n";
11    cout<<"*****\n";
12    return 0;
13 }
```

Process exited after 0.03942 seconds with return value 0
Press any key to continue . . .

5.3 TASK 3

Write a C++ program to print the following:

Expected Output

```
#####
#   My Name is ABC   #
#####
#|   Inter Marks   | 900 |#
#|   Matric Marks   | 900 |#
#####
#|   Total Marks   | 1800 |#
#####
```

```
1  #include <iostream>
2  using namespace std;
3  int main ()
4  {
5      cout<<"#####\n";
6      cout<<"#   My Name is ABC   #\n";
7      cout<<"#####\n";
8      cout<<"#|   Inter Marks   | 900 |#\n";
9      cout<<"#|   Matric Marks   | 900 |#\n";
10     cout<<"#####\n";
11     cout<<"#|   Total Marks   | 1800 |#\n";
12     cout<<"#####\n";
13
14     return 0;
15 }
```

```
C:\Users\mazen\OneDri x + v
#####
#   My Name is ABC   #
#####
#|   Inter Marks   | 900 |#
#|   Matric Marks   | 900 |#
#####
#|   Total Marks   | 1800 |#
#####

Process exited after 0.03897 seconds with return value 0
Press any key to continue . . .
```

5.4 TASK 4

Write a C++ program to print the following:

Expected Output

Inventory Management System				
Number of Item	Name of Item	Quantity	Price	Stock Requirement
1	Transistor	56	\$40	13
2	Diode	45	\$12	16
3	Breadboard	43	\$17	32
4	Wires	100	\$23	23
5	Multimeter	15	\$34	5

```

1 #include <iostream>
2 using namespace std;
3 int main ()
4 {
5     cout<<"***Inventory Management System***\n";
6     cout<<"\n";
7     cout<<"***Inventory Management System***\n";
8     cout<<"\n";
9     cout<<"Number of Item | Name of Item | Quantity | Price | Stock Requirement\n";
10    cout<<"\n";
11    cout<<"1 | Transistor | 56 | $40 | 13\n";
12    cout<<"\n";
13    cout<<"2 | Diode | 45 | $12 | 16\n";
14    cout<<"\n";
15    cout<<"3 | Breadboard | 43 | $17 | 32\n";
16    cout<<"\n";
17    cout<<"4 | Wires | 100 | $23 | 23\n";
18    cout<<"\n";
19    cout<<"5 | Multimeter | 15 | $34 | 5\n";
20    cout<<"\n";
21    return 0;
22 }

```

Number of Item	Name of Item	Quantity	Price	Stock Requirement
1	Transistor	56	\$40	13
2	Diode	45	\$12	16
3	Breadboard	43	\$17	32
4	Wires	100	\$23	23
5	Multimeter	15	\$34	5

Process exited after 0.07206 seconds with return value 0
Press any key to continue . . .

6 HOME ASSIGNMENT

6.1 TASK 1

Total Purchase A customer in a store is purchasing five items. The prices of the five items are:

Price of item 1 = \$12.95

Price of item 2 = \$24.95

Price of item 3 = \$6.95

Price of item 4 = \$14.95

Price of item 5 = \$3.95

Write a program that holds the prices of the five items in five variables. Display each items price, the subtotal of the sale, the amount of sales tax, and the total. Assume the sales tax is 6%.

```

1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     // Prices of items
6     double item1 = 12.95;
7     double item2 = 24.95;
8     double item3 = 6.95;
9     double item4 = 14.95;
10    double item5 = 3.95;
11
12    // calculate subtotal
13    double subtotal = item1 + item2 + item3 + item4 + item5;
14
15    // Sales tax (6%)
16    double salesTax = subtotal * 0.06;
17
18    // Total
19    double total = subtotal + salesTax;
20
21    // Output
22    cout << "Price of item 1 = $" << item1 << endl;
23    cout << "Price of item 2 = $" << item2 << endl;
24    cout << "Price of item 3 = $" << item3 << endl;
25    cout << "Price of item 4 = $" << item4 << endl;
26    cout << "Price of item 5 = $" << item5 << endl;
27
28    cout << "\nSubtotal = $" << subtotal << endl;
29    cout << "Sales Tax (6%) = $" << salesTax << endl;
30    cout << "Total = $" << total << endl;
31
32    return 0;
33 }

```

Price of item 1 = \$12.95
Price of item 2 = \$24.95
Price of item 3 = \$6.95
Price of item 4 = \$14.95
Price of item 5 = \$3.95

Subtotal = \$63.75
Sales Tax (6%) = \$3.825
Total = \$67.575

Process exited after 0.03935 seconds with return value 0
Press any key to continue . . .

6.2 TASK 2

Write a C++ program to print the following:

Expected Output

****Table of 2****

```

2 | x | 1 | = | 2 |
2 | x | 2 | = | 4 |
2 | x | 3 | = | 6 |
2 | x | 4 | = | 8 |
2 | x | 5 | = | 10 |
2 | x | 6 | = | 12 |

```

```

1  #include <iostream>
2  using namespace std;
3  int main ()
4  {
5      cout<<" ****Table of 2**** \n";
6      cout<<" \n";
7      cout<<" 2 | x | 1 | = | 2 | \n";
8      cout<<" 2 | x | 2 | = | 4 | \n";
9      cout<<" 2 | x | 3 | = | 6 | \n";
10     cout<<" 2 | x | 4 | = | 8 | \n";
11     cout<<" 2 | x | 5 | = | 10 | \n";
12     cout<<" 2 | x | 6 | = | 12 | \n";
13     cout<<" \n";
14     return 0;
15 }

```

```

****Table of 2****
2 | x | 1 | = | 2 |
2 | x | 2 | = | 4 |
2 | x | 3 | = | 6 |
2 | x | 4 | = | 8 |
2 | x | 5 | = | 10 |
2 | x | 6 | = | 12 |

-----
Process exited after 2.64 seconds with return value 0
Press any key to continue . . .

```

6.3 TASK 2

Write a C++ program to print the following:

Expected Output

Table of 2 |*| ** Table of 3**

```

** |2|x|1| = |2| |*| |3|x|1| = |3| **
** |2|x|2| = |4| |*| |3|x|2| = |6| **
** |2|x|3| = |6| |*| |3|x|3| = |9| **
** |2|x|4| = |8| |*| |3|x|4| = |12| **

```

```

1  #include <iostream>
2  using namespace std;
3  int main ()
4  {
5      cout<<" **Table of 2** |*| ** Table of 3 ** \n";
6      cout<<" \n";
7      cout<<" ** |2|x|1| = |2| |*| |3|x|1| = |3| ** \n";
8      cout<<" ** |2|x|2| = |4| |*| |3|x|2| = |6| ** \n";
9      cout<<" ** |2|x|3| = |6| |*| |3|x|3| = |9| ** \n";
10     cout<<" ** |2|x|4| = |8| |*| |3|x|4| = |12| ** \n";
11     cout<<" \n";
12     return 0;
13 }

```

```

**Table of 2** |*| ** Table of 3 **
** |2|x|1| = |2| |*| |3|x|1| = |3| **
** |2|x|2| = |4| |*| |3|x|2| = |6| **
** |2|x|3| = |6| |*| |3|x|3| = |9| **
** |2|x|4| = |8| |*| |3|x|4| = |12| **

-----
Process exited after 0.85433 seconds with return value 0
Press any key to continue . . .

```

6.4 TASK 3

Write a C++ program to print the following:

Expected Output

```
-----
|                2+3 |
|                =5  |
|                -----
```

```
| [7] [8] [9] ||  [+]  [-]  |
| [4] [5] [6] ||  [x]  [/]  |
| [1] [2] [3] || [Compute] |
| [A] [0] [B] || [ Clear] |
|                -----
```

In the displayed figure, 2,3, and 5 are integer values which are inserted by the user.

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int num1, num2, result;
6
7      cout<<"Enter first number:\n";
8      cin >> num1;
9      cout<<"Enter second number:\n";
10     cin >> num2;
11
12     result = num1 + num2;
13
14     cout << "-----\n";
15     cout << "          " << num1 << "+" << num2 << " \n";
16     cout << "          " << result << " \n";
17     cout << "-----\n";
18     cout << "| [7] [8] [9] ||  [+]  [-]  |\n";
19     cout << "| [4] [5] [6] ||  [x]  [/]  |\n";
20     cout << "| [1] [2] [3] || [Compute] |\n";
21     cout << "| [A] [0] [B] || [ Clear] |\n";
22     cout << "-----\n";
23
24     return 0;
25 }
```

```
Enter first number:
23
Enter second number:
31
-----
|                23+31 |
|                =54  |
|                -----
| [7] [8] [9] ||  [+]  [-]  |
| [4] [5] [6] ||  [x]  [/]  |
| [1] [2] [3] || [Compute] |
| [A] [0] [B] || [ Clear] |
|                -----

Process exited after 5.573 seconds with return value 0
Press any key to continue . . .
```